

Prepare: FNRP §3.4; review Lesson 09

Primary objective. Approximate the epithermal flux spectrum in an infinite, homogeneous moderator–fuel mixture with resonance absorption.

1 A Quick Review

Recall, we're balancing neutrons at an energy E in an infinite, homogeneous material using the *spectrum equation*:

$$\overbrace{\Sigma_t(E)\phi(E)}^{\text{all losses from } E} = \overbrace{\int_E^\infty \frac{\Sigma_s(E' \rightarrow E)\phi(E')}{E'} dE'}^{\text{gains by scattering}} + \overbrace{\chi(E)s_f'''}^{\text{gains from fission}}.$$

If we *assume* that (1) any neutron above 0.1 MeV scatters immediately to below 0.1 MeV and (2) neutrons below 0.1 MeV can only scatter elastically and isotropically (in the CM system), then

$$\phi(E) \approx \begin{cases} \frac{\chi(E)s_f'''}{\Sigma_t(E)}, & E > 0.1 \text{ MeV}, \\ \frac{s_f'''}{\xi \Sigma_s(E)E}, & 1 \text{ eV} < E \leq 0.1 \text{ MeV}. \end{cases}$$

If $\Sigma_t(E)$ is constant, then $\phi(E) \propto \chi(E)$ in the fast range, while $\phi(E) \propto 1/E$ in the epithermal range.

Checkpoint

What is s_f''' ? What is ξ ? Why is 0.1 MeV significant?

2 Add fuel and absorption

Let m denote the moderator and f the fuel. Define the in-scatter contribution for $i \in \{m, f\}$ as

$$I_i(E) = \int_E^{E/\alpha_i} \frac{\Sigma_s^i(E')\phi(E')}{(1 - \alpha_i)E'} dE'.$$

The epithermal balance becomes

$$\Sigma_t(E)\phi(E) = I_m(E) + I_f(E),$$

where $\Sigma_t = \Sigma_s^m + \Sigma_s^f + \Sigma_a^m + \Sigma_a^f$.

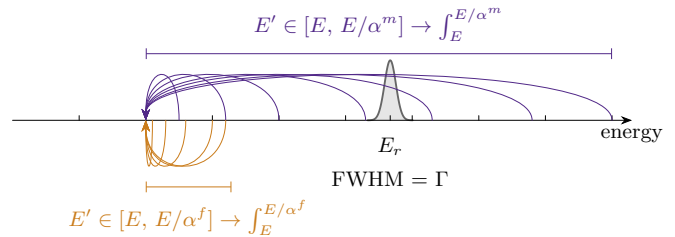
Checkpoint

Why does the left side contain Σ_t , but the right side contains only scattering cross sections and no source?

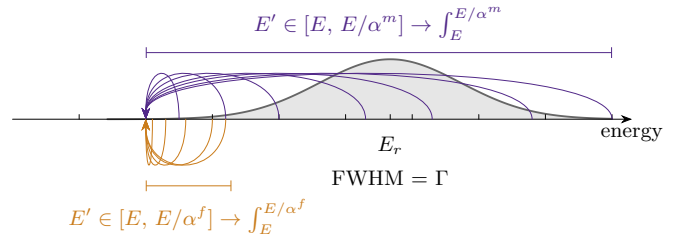
3 Resonances and Absorption

Recall that resonances in the absorption cross section are narrow peaks at specific energies. Clearly, if a neutron *lands* in a resonance it is absorbed, but how likely is it to actually hit that narrow energy window? It depends a lot on its last collision.

Narrow resonance (NR)



Wide resonance (WR)



For a neutron incident near the resonance energy $E \approx E_r$,

$$\alpha_i E_r \leq E' \leq E_r, \quad \Delta E_i = (1 - \alpha_i)E_r.$$

Remember that ΔE_i is the **maximum allowed loss** in a single collision for nuclide i , but it can be much smaller depending on the scattering angle.

Key Assumptions for Narrow Resonance Approximation

The resonance is narrow on the moderator *and* fuel energy-transfer scales:

$$\Delta E_{\text{res}} \ll (1 - \alpha_m)E_r, \quad \Delta E_{\text{res}} \ll (1 - \alpha_f)E_r.$$

Checkpoint

If $E_r = 10 \text{ eV}$ and $\Delta E_{\text{res}} = 0.50 \text{ eV}$, compute ΔE for U-238 and H-1. Which target clearly fails the NR scale comparison?

4 Approximate the scattering source

To evaluate I_m and I_f without first knowing the detailed flux, we assume the following:

1. Absorption is negligible *between* resonances, so the flux has the shape C/E' .
2. $\Sigma_s^i(E) \approx \Sigma_{s,0}^i$ is constant.
3. The resonance is narrow as defined above.

The approximation is made on the right side

Approximate $\Sigma_s^i(E')\phi(E')$ in $I_i(E)$ by $\Sigma_{s,0}^i C/E'$, but **retain the energy-dependent total $\Sigma_t(E)$ on the left.**

5 Derive the NR spectrum

For either scattering nuclide, insert the background approximation from Section 4:

$$\begin{aligned} I_i(E) &\approx \frac{C\Sigma_{s,0}^i}{1-\alpha_i} \int_E^{E/\alpha_i} \frac{dE'}{E'^2} \\ &= \frac{C\Sigma_{s,0}^i}{1-\alpha_i} \left(\frac{1}{E} - \frac{1}{E/\alpha_i} \right) \\ &= \frac{C\Sigma_{s,0}^i}{1-\alpha_i} \left(\frac{1}{E} - \alpha_i \right) \end{aligned}$$

For hydrogen, take $\alpha_i = 0$ and integrate to ∞ ; the same result follows.

Put the two contributions back into balance

Define $\Sigma_{s,0} = \Sigma_{s,0}^m + \Sigma_{s,0}^f$. Then

$$\Sigma_t(E)\phi_{\text{NR}}(E) \approx \frac{C\Sigma_{s,0}}{1-\alpha_i} \left(\frac{1}{E} - \alpha_i \right)$$

Divide by $\Sigma_t(E)$:

$$\phi_{\text{NR}}(E) \approx \frac{C\Sigma_{s,0}}{\Sigma_t(E)E} \implies \phi_{\text{NR}} \propto \frac{1}{\Sigma_t E}$$

This is the NR result (like FNRP 3.29).

Keep the constants straight. $\Sigma_{s,0}$ is the constant background used in the source approximation, *not* the detailed $\Sigma_s(E)$. The slowing-down supply fixes C ; this local argument determines the shape.

Two limiting checks

With the same C , what happens when:

1. absorption vanishes and $\Sigma_t = \Sigma_{s,0}$?
2. $\Sigma_t(E_r)$ increases while the approximate scattering-in source stays fixed?

When NR Fails

When the resonance is not narrow for fuel scattering, the replacement of I_f is not justified by this argument. In such cases, the *wide resonance* (WR) approximation may be more appropriate.

6 Interpret the flux and reaction rate

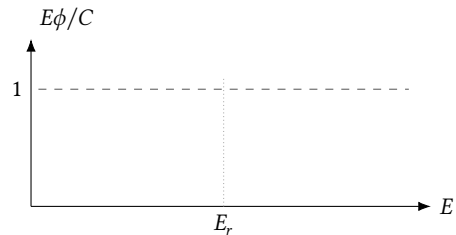
Compare the NR spectrum with the same background $\phi_{\text{bg}} = C/E$:

$$\frac{\phi_{\text{NR}}(E)}{\phi_{\text{bg}}(E)} \approx \frac{\Sigma_{s,0}}{\Sigma_t(E)}$$

Thus a resonance peak in Σ_t produces a depression in the flux relative to its $1/E$ background.

Sketch the consequence

Sketch $E\phi_{\text{NR}}/C$ for an isolated absorption resonance. The dashed line is the undisturbed $1/E$ spectrum in these coordinates.



Schematic prompt; no evaluated nuclear data are implied.

A derived interpretation. A lower flux at the absorption peak reduces absorption relative to using an undisturbed flux there. This is *energy self-shielding*.

A flux dip does not mean zero absorption

Hold fixed the background slowing-down density q supplying the resonance, in neutrons/cm³-s. Using $C = q/(\xi\Sigma_{s,0})$, we obtain

$$\phi_{\text{bg}}(E) = \frac{q}{\xi\Sigma_{s,0}E}, \quad \phi_{\text{NR}}(E) \approx \frac{q}{\xi\Sigma_t(E)E}$$

With $R_a(E) = \Sigma_a(E)\phi_{\text{NR}}(E)$, this gives

$$\frac{\phi_{\text{NR}}}{\phi_{\text{bg}}} \approx \frac{1}{1+x}, \quad \frac{\xi ER_a}{q} \approx \frac{x}{1+x}$$

x	$\phi_{\text{NR}}/\phi_{\text{bg}}$	$\xi ER_a/q$
0	1	0
1	_____	_____
9	_____	_____

As x increases at fixed E , the flux decreases, but the absorption rate approaches $q/(\xi E)$ rather than zero. The last column is a scaled rate, not the fraction of source neutrons absorbed across the resonance.